

PES UNIVERSITY, BENGALURU

Department of Computer Science and Engineering
B.Tech. Computer Science and Engineering | V Semester
UE24CS341A — SOFTWARE ENGINEERING (JACKFRUIT)

PROJECT TEST PLAN

ChargeSlot — EV Charging Station Finder with Slot Dashboard and Auto-Booking System

Team Number: 19 | Project ID: 37 | Team Name: NoteWare

Member	SRN	Owned functionality
Vaibhav Tulsian	PES2UG24AM176	User Accounts, Vehicle Profile and Access Control
Atharva Ravi Jamakhandi	PES2UG24AM196	Station Discovery, Navigation and Real-Time Slot Dashboard
Velkur Tanisha Reddy	PES2UG24AM181	Battery-Aware Booking and Auto-Booking
Vaishnavi Singh	PES2UG24AM177	Payments, Billing and Usage Analytics

Document version: 0.1 (proposed / pre-implementation) | 1 October 2026
Source: ChargeSlot SRS v1.0 (5 September 2026) and Jackfruit SE Guidelines 2026.

Revision History

Name	Date	Reason for Changes	Version
Team NoteWare	1 October 2026	Initial planning draft; implementation and validation pending	0.1

1. Introduction

1.1 Purpose — This is a planning deliverable only. No ChargeSlot software has been implemented; no tests have been executed. Reissue as an execution report after code, test environment, and evidence exist.

1.2 Scope — All 113 SRS functional requirements across the four owned modules. The existing SRS traceability identifiers TC-Fn-nn and STC-Fn-nn are preserved exactly; additional variant IDs use a suffix and do not replace the original IDs.

1.3 Source hierarchy — ChargeSlot SRS v1.0 Appendix C defines identifiers and ownership; Jackfruit Phase-3 guideline defines the nine prescribed columns. The guideline example UT_01 is illustrative; the committed SRS identifiers take precedence.

2. Test Strategy and Levels

Level	Approach	Evidence after implementation
Unit	pytest on pure validators, calculators, scoring, permission helpers; deterministic fixtures and boundary partitions	CI logs, assertion output, branch coverage
Integration	FastAPI routes with PostgreSQL/PostGIS, Redis, gateway mocks and OCPP simulator; concurrency and failure injection	API traces, database state, simulator logs
System	React Native end-to-end driver/operator/admin journeys, hosted Razorpay test mode, Maps stubs and push notification mocks	Timestamped screen recordings, request logs, defect links

3. Test Environment

React Native TypeScript (Android 10+/iOS 15+); Python FastAPI; PostgreSQL/PostGIS; Redis; JWT and Argon2/bcrypt; OCPP 1.6J charge point simulator; Google Maps stub/sandbox; Razorpay test mode; Firebase Cloud Messaging test configuration; pytest, Docker, Jenkins and SonarQube. Environment not provisioned or verified yet.

4. Entry and Exit Criteria by Level

Level	Entry criteria	Exit criteria
Unit	Feature branch merged only after review; functions import; fixture and requirement mapped	All planned unit assertions executed, defects triaged, branch coverage recorded; no invented percentage threshold

Level	Entry criteria	Exit criteria
Integration	Versioned route contracts frozen; PostgreSQL migration and Redis ready; simulator/gateway mock configured	All planned integration cases executed; blocking interface failures resolved; retry/idempotency cases evidenced
System	Integrated build deployed; test accounts for Driver, Operator, Admin; seeded stations and gateway sandbox	All planned system cases executed, no unresolved critical defects, evidence linked, requirement matrix updated

5. Defect Severity and Triage

Severity	Definition	Release disposition
Critical	Account takeover, payment duplication, cross-tenant data disclosure, double booking	Block release and demonstration of affected flow
High	Core registration, discovery, booking or settlement unavailable	Block affected acceptance criterion
Medium	Feature works with material limitation or recoverable incorrect state	Fix or document faculty-approved waiver
Low	Cosmetic or non-blocking wording/layout problem	Track and prioritize

6. Test Ownership and Responsibilities

Member	Module	Responsibility
Vaibhav Tulsian	User Accounts, Vehicle Profile and Access Control	Author fixtures, unit tests, integration contracts and manual system evidence; peer-review one other module
Atharva Ravi Jamakhandi	Station Discovery, Navigation and Real-Time Slot Dashboard	Author fixtures, unit tests, integration contracts and manual system evidence; peer-review one other module
Velkur Tanisha Reddy	Battery-Aware Booking and Auto-Booking	Author fixtures, unit tests, integration contracts and manual system evidence; peer-review one other module
Vaishnavi Singh	Payments, Billing and Usage Analytics	Author fixtures, unit tests, integration contracts and manual system evidence; peer-review one other module

7. Risks and Mitigation

No implementation: every actual outcome remains pending. Simulator fidelity: validate gateway adapter against OCPP 1.6J fixture messages. Race conditions: PostgreSQL transactions and parallel request tests. Payment sandbox: signature checks and idempotency with deterministic replay. Third-party Maps limits: cache, mocks and failure injection. Privacy/RBAC: negative tests with wrong roles and foreign object IDs. Unfixed calendar: capacity assessment is conditional, not a claimed deadline.

8. Planned Test Cases

The following nine headers are reproduced verbatim from the Phase-3 template. “Unit / Integration” indicates a case that must be assigned to one execution suite before running; it is counted as Unit / Integration in planning summaries. Every Actual Result and Test Result is unexecuted.

8.1 User Accounts, Vehicle Profile and Access Control

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-01	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Register account (REQ-F1-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-01: Register account; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-01	F1 — User Accounts, Vehicle Profile and Access Control	System: Register account (REQ-F1-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-01: Register account; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-02	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Password policy (REQ-F1-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-02: Password policy; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-02	F1 — User Accounts, Vehicle Profile and Access Control	System: Password policy (REQ-F1-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-02: Password policy; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-03	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Verify contact details (REQ-F1-03)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-03: Verify contact details; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-03	F1 — User Accounts, Vehicle Profile and Access Control	System: Verify contact details (REQ-F1-03)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-03: Verify contact details; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-04	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Login and tokens (REQ-F1-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-04: Login and tokens; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-04	F1 — User Accounts, Vehicle Profile and Access Control	System: Login and tokens (REQ-F1-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-04: Login and tokens; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-05	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Login lockout (REQ-F1-05)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-05: Login lockout; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-05	F1 — User Accounts, Vehicle Profile and Access Control	System: Login lockout (REQ-F1-05)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-05: Login lockout; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-06	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Password reset (REQ-F1-06)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-06: Password reset; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-06	F1 — User Accounts, Vehicle Profile and Access Control	System: Password reset (REQ-F1-06)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-06: Password reset; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-07	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Logout (REQ-F1-07)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-07: Logout; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-07	F1 — User Accounts, Vehicle Profile and Access Control	System: Logout (REQ-F1-07)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-07: Logout; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-08	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Reject duplicate account (REQ-F1-08)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-08: Reject duplicate account; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-08	F1 — User Accounts, Vehicle Profile and Access Control	System: Reject duplicate account (REQ-F1-08)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-08: Reject duplicate account; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-09	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Generic login error (REQ-F1-09)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-09: Generic login error; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-09	F1 — User Accounts, Vehicle Profile and Access Control	System: Generic login error (REQ-F1-09)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-09: Generic login error; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-10	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Assign one role (REQ-F1-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-10: Assign one role; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-10	F1 — User Accounts, Vehicle Profile and Access Control	System: Assign one role (REQ-F1-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-10: Assign one role; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-11	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Server-side RBAC (REQ-F1-11)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-11: Server-side RBAC; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F1-11	F1 — User Accounts, Vehicle Profile and Access Control	System: Server-side RBAC (REQ-F1-11)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-11: Server-side RBAC; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-12	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Operator data scope (REQ-F1-12)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-12: Operator data scope; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-12	F1 — User Accounts, Vehicle Profile and Access Control	System: Operator data scope (REQ-F1-12)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-12: Operator data scope; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-13	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Admin user management (REQ-F1-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-13: Admin user management; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-13	F1 — User Accounts, Vehicle Profile and Access Control	System: Admin user management (REQ-F1-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-13: Admin user management; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-14	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Audit user actions (REQ-F1-14)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-14: Audit user actions; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-14	F1 — User Accounts, Vehicle Profile and Access Control	System: Audit user actions (REQ-F1-14)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-14: Audit user actions; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-15	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Operator onboarding (REQ-F1-15)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-15: Operator onboarding; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-15	F1 — User Accounts, Vehicle Profile and Access Control	System: Operator onboarding (REQ-F1-15)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-15: Operator onboarding; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-16	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Edit profile and preferences (REQ-F1-16)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-16: Edit profile and preferences; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-16	F1 — User Accounts, Vehicle Profile and Access Control	System: Edit profile and preferences (REQ-F1-16)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-16: Edit profile and preferences; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-17	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Profile validation (REQ-F1-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-17: Profile validation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-17	F1 — User Accounts, Vehicle Profile and Access Control	System: Profile validation (REQ-F1-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-17: Profile validation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-18	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Manual location fallback (REQ-F1-18)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-18: Manual location fallback; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-18	F1 — User Accounts, Vehicle Profile and Access Control	System: Manual location fallback (REQ-F1-18)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-18: Manual location fallback; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-19	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Location retention limit (REQ-F1-19)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-19: Location retention limit; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-19	F1 — User Accounts, Vehicle Profile and Access Control	System: Location retention limit (REQ-F1-19)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-19: Location retention limit; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-20	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Account deletion (REQ-F1-20)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-20: Account deletion; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-20	F1 — User Accounts, Vehicle Profile and Access Control	System: Account deletion (REQ-F1-20)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-20: Account deletion; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-21	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Vehicle CRUD (REQ-F1-21)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-21: Vehicle CRUD; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-21	F1 — User Accounts, Vehicle Profile and Access Control	System: Vehicle CRUD (REQ-F1-21)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-21: Vehicle CRUD; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-22	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Vehicle limit and default (REQ-F1-22)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-22: Vehicle limit and default; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F1-22	F1 — User Accounts, Vehicle Profile and Access Control	System: Vehicle limit and default (REQ-F1-22)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-22: Vehicle limit and default; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-23	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Preset EV models (REQ-F1-23)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-23: Preset EV models; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-23	F1 — User Accounts, Vehicle Profile and Access Control	System: Preset EV models (REQ-F1-23)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-23: Preset EV models; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-24	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Vehicle value validation (REQ-F1-24)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-24: Vehicle value validation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-24	F1 — User Accounts, Vehicle Profile and Access Control	System: Vehicle value validation (REQ-F1-24)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-24: Vehicle value validation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-25	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Protect booked vehicle (REQ-F1-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-25: Protect booked vehicle; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-25	F1 — User Accounts, Vehicle Profile and Access Control	System: Protect booked vehicle (REQ-F1-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-25: Protect booked vehicle; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-26	F1 — User Accounts, Vehicle Profile and Access Control	Unit / Integration: Driver sees own data only (REQ-F1-26)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-26: Driver sees own data only; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F1-26	F1 — User Accounts, Vehicle Profile and Access Control	System: Driver sees own data only (REQ-F1-26)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F1-26: Driver sees own data only; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F1-17-V1	F1 — User Accounts, Vehicle Profile and Access Control	Unit: Invalid or missing mandatory input (REQ-F1-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	missing field, malformed string, negative numeric, unknown ID	Reject invalid payload; no database mutation.	Not executed — pending implementation	Pending
TC-F1-17-V2	F1 — User Accounts, Vehicle Profile and Access Control	Unit: Boundary value at documented limit (REQ-F1-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	0, 1, max-1, max, max+1 where SRS specifies limit	Accept only values inside SRS-defined limits; reject out-of-range values.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F1-10-V3	F1 — User Accounts, Vehicle Profile and Access Control	Unit: Wrong-role access must be denied (REQ-F1-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	Driver token accessing operator/admin endpoint; foreign resource ID	403 Forbidden (or 404 to avoid resource disclosure); no mutation or data leak.	Not executed — pending implementation	Pending
TC-F1-01-V4	F1 — User Accounts, Vehicle Profile and Access Control	Integration: Upstream dependency unavailable / timeout (REQ-F1-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	gateway timeout / Redis unavailable / stale connector status	Return controlled failure or fallback; no duplicate booking/payment.	Not executed — pending implementation	Pending

8.2 Station Discovery, Navigation and Real-Time Slot Dashboard

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-01	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Station CRUD (REQ-F2-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-01: Station CRUD; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-01	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Station CRUD (REQ-F2-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-01: Station CRUD; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-02	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Charger and connector CRUD (REQ-F2-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-02: Charger and connector CRUD; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-02	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Charger and connector CRUD (REQ-F2-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-02: Charger and connector CRUD; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-03	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Station approval gate (REQ-F2-03)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-03: Station approval gate; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-03	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Station approval gate (REQ-F2-03)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-03: Station approval gate; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-04	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Station validation (REQ-F2-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-04: Station validation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-04	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Station validation (REQ-F2-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-04: Station validation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-05	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Out-of-service marking (REQ-F2-05)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-05: Out-of-service marking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-05	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Out-of-service marking (REQ-F2-05)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-05: Out-of-service marking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-06	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Maintenance windows (REQ-F2-06)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-06: Maintenance windows; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-06	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Maintenance windows (REQ-F2-06)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-06: Maintenance windows; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-07	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Nearby station search (REQ-F2-07)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-07: Nearby station search; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-07	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Nearby station search (REQ-F2-07)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-07: Nearby station search; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-08	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Availability markers (REQ-F2-08)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-08: Availability markers; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-08	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Availability markers (REQ-F2-08)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-08: Availability markers; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-09	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Filter stations (REQ-F2-09)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-09: Filter stations; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-09	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Filter stations (REQ-F2-09)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-09: Filter stations; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-10	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Sort stations (REQ-F2-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-10: Sort stations; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F2-10	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Sort stations (REQ-F2-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-10: Sort stations; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-11	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Text search (REQ-F2-11)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-11: Text search; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-11	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Text search (REQ-F2-11)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-11: Text search; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-12	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Pagination and clustering (REQ-F2-12)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-12: Pagination and clustering; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-12	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Pagination and clustering (REQ-F2-12)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-12: Pagination and clustering; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-13	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Station detail (REQ-F2-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-13: Station detail; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-13	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Station detail (REQ-F2-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-13: Station detail; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-14	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Favourite stations (REQ-F2-14)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-14: Favourite stations; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-14	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Favourite stations (REQ-F2-14)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-14: Favourite stations; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-15	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Rate a station (REQ-F2-15)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-15: Rate a station; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-15	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Rate a station (REQ-F2-15)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-15: Rate a station; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-16	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Report faulty charger (REQ-F2-16)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-16: Report faulty charger; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-16	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Report faulty charger (REQ-F2-16)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-16: Report faulty charger; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-17	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Ingest connector status (REQ-F2-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-17: Ingest connector status; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-17	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Ingest connector status (REQ-F2-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-17: Ingest connector status; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-18	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Live status display (REQ-F2-18)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-18: Live status display; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-18	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Live status display (REQ-F2-18)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-18: Live status display; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-19	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Live status push (REQ-F2-19)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-19: Live status push; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-19	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Live status push (REQ-F2-19)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-19: Live status push; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-20	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Stale status handling (REQ-F2-20)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-20: Stale status handling; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-20	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Stale status handling (REQ-F2-20)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-20: Stale status handling; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-21	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Reserved status (REQ-F2-21)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-21: Reserved status; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F2-21	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Reserved status (REQ-F2-21)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-21: Reserved status; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-22	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Slot generation (REQ-F2-22)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-22: Slot generation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-22	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Slot generation (REQ-F2-22)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-22: Slot generation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-23	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Slot timeline (REQ-F2-23)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-23: Slot timeline; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-23	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Slot timeline (REQ-F2-23)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-23: Slot timeline; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-24	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Estimated wait time (REQ-F2-24)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-24: Estimated wait time; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-24	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Estimated wait time (REQ-F2-24)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-24: Estimated wait time; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-25	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Route preview (REQ-F2-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-25: Route preview; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-25	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Route preview (REQ-F2-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-25: Route preview; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-26	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Navigation hand-off (REQ-F2-26)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-26: Navigation hand-off; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-26	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Navigation hand-off (REQ-F2-26)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-26: Navigation hand-off; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F2-27	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: ETA versus slot check (REQ-F2-27)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-27: ETA versus slot check; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-27	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: ETA versus slot check (REQ-F2-27)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-27: ETA versus slot check; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-28	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit / Integration: Route cache and fallback (REQ-F2-28)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-28: Route cache and fallback; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F2-28	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	System: Route cache and fallback (REQ-F2-28)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F2-28: Route cache and fallback; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F2-04-V1	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit: Invalid or missing mandatory input (REQ-F2-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	missing field, malformed string, negative numeric, unknown ID	Reject invalid payload; no database mutation.	Not executed — pending implementation	Pending
TC-F2-04-V2	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit: Boundary value at documented limit (REQ-F2-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	0, 1, max-1, max, max+1 where SRS specifies limit	Accept only values inside SRS-defined limits; reject out-of-range values.	Not executed — pending implementation	Pending
TC-F2-01-V3	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Unit: Wrong-role access must be denied (REQ-F2-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	Driver token accessing operator/admin endpoint; foreign resource ID	403 Forbidden (or 404 to avoid resource disclosure); no mutation or data leak.	Not executed — pending implementation	Pending
TC-F2-17-V4	F2 — Station Discovery, Navigation and Real-Time Slot Dashboard	Integration: Upstream dependency unavailable / timeout (REQ-F2-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	gateway timeout / Redis unavailable / stale connector status	Return controlled failure or fallback; no duplicate booking/payment.	Not executed — pending implementation	Pending

8.3 Battery-Aware Booking and Auto-Booking

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-01	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Battery level input (REQ-F3-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-01: Battery level input; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-01	F3 — Battery-Aware Booking and Auto-Booking	System: Battery level input (REQ-F3-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-01: Battery level input; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-02	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Battery input validation (REQ-F3-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-02: Battery input validation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-02	F3 — Battery-Aware Booking and Auto-Booking	System: Battery input validation (REQ-F3-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-02: Battery input validation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-03	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Usable range calculation (REQ-F3-03)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-03: Usable range calculation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-03	F3 — Battery-Aware Booking and Auto-Booking	System: Usable range calculation (REQ-F3-03)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-03: Usable range calculation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-04	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Reachability filter (REQ-F3-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-04: Reachability filter; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-04	F3 — Battery-Aware Booking and Auto-Booking	System: Reachability filter (REQ-F3-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-04: Reachability filter; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-05	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Connector compatibility (REQ-F3-05)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-05: Connector compatibility; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-05	F3 — Battery-Aware Booking and Auto-Booking	System: Connector compatibility (REQ-F3-05)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-05: Connector compatibility; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-06	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Energy estimate (REQ-F3-06)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-06: Energy estimate; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-06	F3 — Battery-Aware Booking and Auto-Booking	System: Energy estimate (REQ-F3-06)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-06: Energy estimate; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-07	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Charging time estimate (REQ-F3-07)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-07: Charging time estimate; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F3-07	F3 — Battery-Aware Booking and Auto-Booking	System: Charging time estimate (REQ-F3-07)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-07: Charging time estimate; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-08	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Slot length rule (REQ-F3-08)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-08: Slot length rule; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-08	F3 — Battery-Aware Booking and Auto-Booking	System: Slot length rule (REQ-F3-08)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-08: Slot length rule; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-09	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Cost estimate (REQ-F3-09)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-09: Cost estimate; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-09	F3 — Battery-Aware Booking and Auto-Booking	System: Cost estimate (REQ-F3-09)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-09: Cost estimate; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-10	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Recommended slots (REQ-F3-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-10: Recommended slots; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-10	F3 — Battery-Aware Booking and Auto-Booking	System: Recommended slots (REQ-F3-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-10: Recommended slots; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-11	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Manual slot choice (REQ-F3-11)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-11: Manual slot choice; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-11	F3 — Battery-Aware Booking and Auto-Booking	System: Manual slot choice (REQ-F3-11)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-11: Manual slot choice; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-12	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Slot hold (REQ-F3-12)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-12: Slot hold; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-12	F3 — Battery-Aware Booking and Auto-Booking	System: Slot hold (REQ-F3-12)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-12: Slot hold; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-13	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: No double booking (REQ-F3-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-13: No double booking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-13	F3 — Battery-Aware Booking and Auto-Booking	System: No double booking (REQ-F3-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-13: No double booking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-14	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Driver overlap rule (REQ-F3-14)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-14: Driver overlap rule; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-14	F3 — Battery-Aware Booking and Auto-Booking	System: Driver overlap rule (REQ-F3-14)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-14: Driver overlap rule; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-15	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Confirm booking (REQ-F3-15)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-15: Confirm booking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-15	F3 — Battery-Aware Booking and Auto-Booking	System: Confirm booking (REQ-F3-15)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-15: Confirm booking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-16	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Auto-book action (REQ-F3-16)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-16: Auto-book action; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-16	F3 — Battery-Aware Booking and Auto-Booking	System: Auto-book action (REQ-F3-16)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-16: Auto-book action; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-17	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Weighted slot scoring (REQ-F3-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-17: Weighted slot scoring; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-17	F3 — Battery-Aware Booking and Auto-Booking	System: Weighted slot scoring (REQ-F3-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-17: Weighted slot scoring; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-18	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Explain the choice (REQ-F3-18)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-18: Explain the choice; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F3-18	F3 — Battery-Aware Booking and Auto-Booking	System: Explain the choice (REQ-F3-18)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-18: Explain the choice; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-19	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Confirmation modes (REQ-F3-19)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-19: Confirmation modes; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-19	F3 — Battery-Aware Booking and Auto-Booking	System: Confirmation modes (REQ-F3-19)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-19: Confirmation modes; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-20	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Low-battery trigger (REQ-F3-20)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-20: Low-battery trigger; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-20	F3 — Battery-Aware Booking and Auto-Booking	System: Low-battery trigger (REQ-F3-20)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-20: Low-battery trigger; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-21	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: No-slot handling (REQ-F3-21)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-21: No-slot handling; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-21	F3 — Battery-Aware Booking and Auto-Booking	System: No-slot handling (REQ-F3-21)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-21: No-slot handling; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-22	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Booking state machine (REQ-F3-22)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-22: Booking state machine; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-22	F3 — Battery-Aware Booking and Auto-Booking	System: Booking state machine (REQ-F3-22)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-22: Booking state machine; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-23	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Modify booking (REQ-F3-23)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-23: Modify booking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-23	F3 — Battery-Aware Booking and Auto-Booking	System: Modify booking (REQ-F3-23)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-23: Modify booking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-24	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Cancel booking (REQ-F3-24)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-24: Cancel booking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-24	F3 — Battery-Aware Booking and Auto-Booking	System: Cancel booking (REQ-F3-24)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-24: Cancel booking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-25	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: No-show handling (REQ-F3-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-25: No-show handling; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-25	F3 — Battery-Aware Booking and Auto-Booking	System: No-show handling (REQ-F3-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-25: No-show handling; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-26	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Auto-rebooking (REQ-F3-26)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-26: Auto-rebooking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-26	F3 — Battery-Aware Booking and Auto-Booking	System: Auto-rebooking (REQ-F3-26)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-26: Auto-rebooking; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-27	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Check-in (REQ-F3-27)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-27: Check-in; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-27	F3 — Battery-Aware Booking and Auto-Booking	System: Check-in (REQ-F3-27)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-27: Check-in; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-28	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Remote start (REQ-F3-28)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-28: Remote start; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-28	F3 — Battery-Aware Booking and Auto-Booking	System: Remote start (REQ-F3-28)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-28: Remote start; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-29	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Live session monitor (REQ-F3-29)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-29: Live session monitor; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F3-29	F3 — Battery-Aware Booking and Auto-Booking	System: Live session monitor (REQ-F3-29)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-29: Live session monitor; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-30	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Stop session (REQ-F3-30)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-30: Stop session; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-30	F3 — Battery-Aware Booking and Auto-Booking	System: Stop session (REQ-F3-30)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-30: Stop session; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-31	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Start failure handling (REQ-F3-31)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-31: Start failure handling; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-31	F3 — Battery-Aware Booking and Auto-Booking	System: Start failure handling (REQ-F3-31)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-31: Start failure handling; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-32	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Idle fee (REQ-F3-32)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-32: Idle fee; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-32	F3 — Battery-Aware Booking and Auto-Booking	System: Idle fee (REQ-F3-32)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-32: Idle fee; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-33	F3 — Battery-Aware Booking and Auto-Booking	Unit / Integration: Booking notifications (REQ-F3-33)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-33: Booking notifications; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F3-33	F3 — Battery-Aware Booking and Auto-Booking	System: Booking notifications (REQ-F3-33)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F3-33: Booking notifications; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F3-02-V1	F3 — Battery-Aware Booking and Auto-Booking	Unit: Invalid or missing mandatory input (REQ-F3-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	missing field, malformed string, negative numeric, unknown ID	Reject invalid payload; no database mutation.	Not executed — pending implementation	Pending
TC-F3-02-V2	F3 — Battery-Aware Booking and Auto-Booking	Unit: Boundary value at documented limit (REQ-F3-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	0, 1, max-1, max, max+1 where SRS specifies limit	Accept only values inside SRS-defined limits; reject out-of-range values.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F3-01-V3	F3 — Battery-Aware Booking and Auto-Booking	Unit: Wrong-role access must be denied (REQ-F3-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	Driver token accessing operator/admin endpoint; foreign resource ID	403 Forbidden (or 404 to avoid resource disclosure); no mutation or data leak.	Not executed — pending implementation	Pending
TC-F3-13-V4	F3 — Battery-Aware Booking and Auto-Booking	Integration: Upstream dependency unavailable / timeout (REQ-F3-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	gateway timeout / Redis unavailable / stale connector status	Return controlled failure or fallback; no duplicate booking/payment.	Not executed — pending implementation	Pending

8.4 Payments, Billing and Usage Analytics

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-01	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Payment methods (REQ-F4-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-01: Payment methods; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-01	F4 — Payments, Billing and Usage Analytics	System: Payment methods (REQ-F4-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-01: Payment methods; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-02	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Tokenised payment methods (REQ-F4-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-02: Tokenised payment methods; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-02	F4 — Payments, Billing and Usage Analytics	System: Tokenised payment methods (REQ-F4-02)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-02: Tokenised payment methods; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-03	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Remove payment method (REQ-F4-03)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-03: Remove payment method; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-03	F4 — Payments, Billing and Usage Analytics	System: Remove payment method (REQ-F4-03)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-03: Remove payment method; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-04	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Reservation fee (REQ-F4-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-04: Reservation fee; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-04	F4 — Payments, Billing and Usage Analytics	System: Reservation fee (REQ-F4-04)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-04: Reservation fee; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-05	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Final bill computation (REQ-F4-05)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-05: Final bill computation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-05	F4 — Payments, Billing and Usage Analytics	System: Final bill computation (REQ-F4-05)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-05: Final bill computation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-06	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Tariff snapshot (REQ-F4-06)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-06: Tariff snapshot; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-06	F4 — Payments, Billing and Usage Analytics	System: Tariff snapshot (REQ-F4-06)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-06: Tariff snapshot; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-07	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Settle balance (REQ-F4-07)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-07: Settle balance; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-07	F4 — Payments, Billing and Usage Analytics	System: Settle balance (REQ-F4-07)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-07: Settle balance; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-08	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Refunds (REQ-F4-08)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-08: Refunds; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-08	F4 — Payments, Billing and Usage Analytics	System: Refunds (REQ-F4-08)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-08: Refunds; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-09	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Payment records (REQ-F4-09)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-09: Payment records; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-09	F4 — Payments, Billing and Usage Analytics	System: Payment records (REQ-F4-09)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-09: Payment records; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-10	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Idempotent retry (REQ-F4-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-10: Idempotent retry; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F4-10	F4 — Payments, Billing and Usage Analytics	System: Idempotent retry (REQ-F4-10)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-10: Idempotent retry; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-11	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Webhook verification (REQ-F4-11)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-11: Webhook verification; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-11	F4 — Payments, Billing and Usage Analytics	System: Webhook verification (REQ-F4-11)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-11: Webhook verification; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-12	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Server-side verification (REQ-F4-12)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-12: Server-side verification; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-12	F4 — Payments, Billing and Usage Analytics	System: Server-side verification (REQ-F4-12)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-12: Server-side verification; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-13	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Daily reconciliation (REQ-F4-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-13: Daily reconciliation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-13	F4 — Payments, Billing and Usage Analytics	System: Daily reconciliation (REQ-F4-13)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-13: Daily reconciliation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-14	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Invoice generation (REQ-F4-14)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-14: Invoice generation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-14	F4 — Payments, Billing and Usage Analytics	System: Invoice generation (REQ-F4-14)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-14: Invoice generation; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-15	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Payment history (REQ-F4-15)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-15: Payment history; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-15	F4 — Payments, Billing and Usage Analytics	System: Payment history (REQ-F4-15)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-15: Payment history; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-16	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Tariff management (REQ-F4-16)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-16: Tariff management; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-16	F4 — Payments, Billing and Usage Analytics	System: Tariff management (REQ-F4-16)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-16: Tariff management; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-17	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Operator earnings (REQ-F4-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-17: Operator earnings; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-17	F4 — Payments, Billing and Usage Analytics	System: Operator earnings (REQ-F4-17)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-17: Operator earnings; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-18	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Driver usage summary (REQ-F4-18)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-18: Driver usage summary; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-18	F4 — Payments, Billing and Usage Analytics	System: Driver usage summary (REQ-F4-18)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-18: Driver usage summary; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-19	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Driver trend charts (REQ-F4-19)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-19: Driver trend charts; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-19	F4 — Payments, Billing and Usage Analytics	System: Driver trend charts (REQ-F4-19)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-19: Driver trend charts; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-20	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Top stations and CO2 (REQ-F4-20)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-20: Top stations and CO2; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-20	F4 — Payments, Billing and Usage Analytics	System: Top stations and CO2 (REQ-F4-20)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-20: Top stations and CO2; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-21	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Operator analytics (REQ-F4-21)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-21: Operator analytics; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
STC-F4-21	F4 — Payments, Billing and Usage Analytics	System: Operator analytics (REQ-F4-21)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-21: Operator analytics; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-22	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Platform analytics (REQ-F4-22)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-22: Platform analytics; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-22	F4 — Payments, Billing and Usage Analytics	System: Platform analytics (REQ-F4-22)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-22: Platform analytics; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-23	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Date range filter (REQ-F4-23)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-23: Date range filter; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-23	F4 — Payments, Billing and Usage Analytics	System: Date range filter (REQ-F4-23)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-23: Date range filter; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-24	F4 — Payments, Billing and Usage Analytics	Unit / Integration: CSV export (REQ-F4-24)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-24: CSV export; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-24	F4 — Payments, Billing and Usage Analytics	System: CSV export (REQ-F4-24)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-24: CSV export; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-25	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Role-scoped analytics (REQ-F4-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-25: Role-scoped analytics; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-25	F4 — Payments, Billing and Usage Analytics	System: Role-scoped analytics (REQ-F4-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-25: Role-scoped analytics; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
TC-F4-26	F4 — Payments, Billing and Usage Analytics	Unit / Integration: Analytics refresh (REQ-F4-26)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-26: Analytics refresh; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending
STC-F4-26	F4 — Payments, Billing and Usage Analytics	System: Analytics refresh (REQ-F4-26)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	fixture: valid registered Driver / seeded record; repeat with invalid and boundary partitions	SRS REQ-F4-26: Analytics refresh; enforce documented rules and reject invalid access.	Not executed — pending implementation	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
TC-F4-01-V1	F4 — Payments, Billing and Usage Analytics	Unit: Invalid or missing mandatory input (REQ-F4-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	missing field, malformed string, negative numeric, unknown ID	Reject invalid payload; no database mutation.	Not executed — pending implementation	Pending
TC-F4-01-V2	F4 — Payments, Billing and Usage Analytics	Unit: Boundary value at documented limit (REQ-F4-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	0, 1, max-1, max, max+1 where SRS specifies limit	Accept only values inside SRS-defined limits; reject out-of-range values.	Not executed — pending implementation	Pending
TC-F4-25-V3	F4 — Payments, Billing and Usage Analytics	Unit: Wrong-role access must be denied (REQ-F4-25)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	Driver token accessing operator/admin endpoint; foreign resource ID	403 Forbidden (or 404 to avoid resource disclosure); no mutation or data leak.	Not executed — pending implementation	Pending
TC-F4-01-V4	F4 — Payments, Billing and Usage Analytics	Integration: Upstream dependency unavailable / timeout (REQ-F4-01)	Code deployed; fixtures and correct/incorrect role accounts provisioned.	1. Seed fixture. 2. Submit test payload. 3. Inspect response and persistent state. 4. Compare to SRS.	gateway timeout / Redis unavailable / stale connector status	Return controlled failure or fallback; no duplicate booking/payment.	Not executed — pending implementation	Pending

9. Requirement-to-Test-Case Traceability

The baseline TC and STC identifiers below are taken directly from the SRS matrix; variant cases supplement the baseline coverage.

Requirement ID	Description	Unit / Integration ID	System Test Case ID
REQ-F1-01	Register account	TC-F1-01	STC-F1-01
REQ-F1-02	Password policy	TC-F1-02	STC-F1-02
REQ-F1-03	Verify contact details	TC-F1-03	STC-F1-03
REQ-F1-04	Login and tokens	TC-F1-04	STC-F1-04
REQ-F1-05	Login lockout	TC-F1-05	STC-F1-05
REQ-F1-06	Password reset	TC-F1-06	STC-F1-06
REQ-F1-07	Logout	TC-F1-07	STC-F1-07
REQ-F1-08	Reject duplicate account	TC-F1-08	STC-F1-08
REQ-F1-09	Generic login error	TC-F1-09	STC-F1-09
REQ-F1-10	Assign one role	TC-F1-10	STC-F1-10
REQ-F1-11	Server-side RBAC	TC-F1-11	STC-F1-11
REQ-F1-12	Operator data scope	TC-F1-12	STC-F1-12
REQ-F1-13	Admin user management	TC-F1-13	STC-F1-13
REQ-F1-14	Audit user actions	TC-F1-14	STC-F1-14

Requirement ID	Description	Unit / Integration ID	System Test Case ID
REQ-F1-15	Operator onboarding	TC-F1-15	STC-F1-15
REQ-F1-16	Edit profile and preferences	TC-F1-16	STC-F1-16
REQ-F1-17	Profile validation	TC-F1-17	STC-F1-17
REQ-F1-18	Manual location fallback	TC-F1-18	STC-F1-18
REQ-F1-19	Location retention limit	TC-F1-19	STC-F1-19
REQ-F1-20	Account deletion	TC-F1-20	STC-F1-20
REQ-F1-21	Vehicle CRUD	TC-F1-21	STC-F1-21
REQ-F1-22	Vehicle limit and default	TC-F1-22	STC-F1-22
REQ-F1-23	Preset EV models	TC-F1-23	STC-F1-23
REQ-F1-24	Vehicle value validation	TC-F1-24	STC-F1-24
REQ-F1-25	Protect booked vehicle	TC-F1-25	STC-F1-25
REQ-F1-26	Driver sees own data only	TC-F1-26	STC-F1-26
REQ-F2-01	Station CRUD	TC-F2-01	STC-F2-01
REQ-F2-02	Charger and connector CRUD	TC-F2-02	STC-F2-02
REQ-F2-03	Station approval gate	TC-F2-03	STC-F2-03
REQ-F2-04	Station validation	TC-F2-04	STC-F2-04
REQ-F2-05	Out-of-service marking	TC-F2-05	STC-F2-05
REQ-F2-06	Maintenance windows	TC-F2-06	STC-F2-06
REQ-F2-07	Nearby station search	TC-F2-07	STC-F2-07
REQ-F2-08	Availability markers	TC-F2-08	STC-F2-08
REQ-F2-09	Filter stations	TC-F2-09	STC-F2-09
REQ-F2-10	Sort stations	TC-F2-10	STC-F2-10
REQ-F2-11	Text search	TC-F2-11	STC-F2-11
REQ-F2-12	Pagination and clustering	TC-F2-12	STC-F2-12
REQ-F2-13	Station detail	TC-F2-13	STC-F2-13
REQ-F2-14	Favourite stations	TC-F2-14	STC-F2-14
REQ-F2-15	Rate a station	TC-F2-15	STC-F2-15
REQ-F2-16	Report faulty charger	TC-F2-16	STC-F2-16

Requirement ID	Description	Unit / Integration ID	System Test Case ID
REQ-F2-17	Ingest connector status	TC-F2-17	STC-F2-17
REQ-F2-18	Live status display	TC-F2-18	STC-F2-18
REQ-F2-19	Live status push	TC-F2-19	STC-F2-19
REQ-F2-20	Stale status handling	TC-F2-20	STC-F2-20
REQ-F2-21	Reserved status	TC-F2-21	STC-F2-21
REQ-F2-22	Slot generation	TC-F2-22	STC-F2-22
REQ-F2-23	Slot timeline	TC-F2-23	STC-F2-23
REQ-F2-24	Estimated wait time	TC-F2-24	STC-F2-24
REQ-F2-25	Route preview	TC-F2-25	STC-F2-25
REQ-F2-26	Navigation hand-off	TC-F2-26	STC-F2-26
REQ-F2-27	ETA versus slot check	TC-F2-27	STC-F2-27
REQ-F2-28	Route cache and fallback	TC-F2-28	STC-F2-28
REQ-F3-01	Battery level input	TC-F3-01	STC-F3-01
REQ-F3-02	Battery input validation	TC-F3-02	STC-F3-02
REQ-F3-03	Usable range calculation	TC-F3-03	STC-F3-03
REQ-F3-04	Reachability filter	TC-F3-04	STC-F3-04
REQ-F3-05	Connector compatibility	TC-F3-05	STC-F3-05
REQ-F3-06	Energy estimate	TC-F3-06	STC-F3-06
REQ-F3-07	Charging time estimate	TC-F3-07	STC-F3-07
REQ-F3-08	Slot length rule	TC-F3-08	STC-F3-08
REQ-F3-09	Cost estimate	TC-F3-09	STC-F3-09
REQ-F3-10	Recommended slots	TC-F3-10	STC-F3-10
REQ-F3-11	Manual slot choice	TC-F3-11	STC-F3-11
REQ-F3-12	Slot hold	TC-F3-12	STC-F3-12
REQ-F3-13	No double booking	TC-F3-13	STC-F3-13
REQ-F3-14	Driver overlap rule	TC-F3-14	STC-F3-14
REQ-F3-15	Confirm booking	TC-F3-15	STC-F3-15
REQ-F3-16	Auto-book action	TC-F3-16	STC-F3-16

Requirement ID	Description	Unit / Integration ID	System Test Case ID
REQ-F3-17	Weighted slot scoring	TC-F3-17	STC-F3-17
REQ-F3-18	Explain the choice	TC-F3-18	STC-F3-18
REQ-F3-19	Confirmation modes	TC-F3-19	STC-F3-19
REQ-F3-20	Low-battery trigger	TC-F3-20	STC-F3-20
REQ-F3-21	No-slot handling	TC-F3-21	STC-F3-21
REQ-F3-22	Booking state machine	TC-F3-22	STC-F3-22
REQ-F3-23	Modify booking	TC-F3-23	STC-F3-23
REQ-F3-24	Cancel booking	TC-F3-24	STC-F3-24
REQ-F3-25	No-show handling	TC-F3-25	STC-F3-25
REQ-F3-26	Auto-rebooking	TC-F3-26	STC-F3-26
REQ-F3-27	Check-in	TC-F3-27	STC-F3-27
REQ-F3-28	Remote start	TC-F3-28	STC-F3-28
REQ-F3-29	Live session monitor	TC-F3-29	STC-F3-29
REQ-F3-30	Stop session	TC-F3-30	STC-F3-30
REQ-F3-31	Start failure handling	TC-F3-31	STC-F3-31
REQ-F3-32	Idle fee	TC-F3-32	STC-F3-32
REQ-F3-33	Booking notifications	TC-F3-33	STC-F3-33
REQ-F4-01	Payment methods	TC-F4-01	STC-F4-01
REQ-F4-02	Tokenised payment methods	TC-F4-02	STC-F4-02
REQ-F4-03	Remove payment method	TC-F4-03	STC-F4-03
REQ-F4-04	Reservation fee	TC-F4-04	STC-F4-04
REQ-F4-05	Final bill computation	TC-F4-05	STC-F4-05
REQ-F4-06	Tariff snapshot	TC-F4-06	STC-F4-06
REQ-F4-07	Settle balance	TC-F4-07	STC-F4-07
REQ-F4-08	Refunds	TC-F4-08	STC-F4-08
REQ-F4-09	Payment records	TC-F4-09	STC-F4-09
REQ-F4-10	Idempotent retry	TC-F4-10	STC-F4-10
REQ-F4-11	Webhook verification	TC-F4-11	STC-F4-11

Requirement ID	Description	Unit / Integration ID	System Test Case ID
REQ-F4-12	Server-side verification	TC-F4-12	STC-F4-12
REQ-F4-13	Daily reconciliation	TC-F4-13	STC-F4-13
REQ-F4-14	Invoice generation	TC-F4-14	STC-F4-14
REQ-F4-15	Payment history	TC-F4-15	STC-F4-15
REQ-F4-16	Tariff management	TC-F4-16	STC-F4-16
REQ-F4-17	Operator earnings	TC-F4-17	STC-F4-17
REQ-F4-18	Driver usage summary	TC-F4-18	STC-F4-18
REQ-F4-19	Driver trend charts	TC-F4-19	STC-F4-19
REQ-F4-20	Top stations and CO2	TC-F4-20	STC-F4-20
REQ-F4-21	Operator analytics	TC-F4-21	STC-F4-21
REQ-F4-22	Platform analytics	TC-F4-22	STC-F4-22
REQ-F4-23	Date range filter	TC-F4-23	STC-F4-23
REQ-F4-24	CSV export	TC-F4-24	STC-F4-24
REQ-F4-25	Role-scoped analytics	TC-F4-25	STC-F4-25
REQ-F4-26	Analytics refresh	TC-F4-26	STC-F4-26

10. Test Inventory Summary

Module	SRS requirements	Unit / Integration baseline	System baseline	Additional variants	Total planned	Executed	Pending
F1	26	26	26	4	56	0	56
F2	28	28	28	4	60	0	60
F3	33	33	33	4	70	0	70
F4	26	26	26	4	56	0	56
TOTAL	113	113	113	16	242	0	242

Execution status: zero executed. A separate revision must replace all pending fields with witnessed actual results, pass/fail decisions, environment identifiers, defect IDs and evidence links.